

Surya Pavan Peruri

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Education

Master of Science, *University of Houston*

Jan 2023 – Dec 2024 | Houston, USA

Engineering Data Science

Bachelor of Technology, *Vellore Institute of Technology*

July 2017 – June 2021 | Vellore, India

Computer Science and Engineering

Technical skills

- **Programming Languages:** Python, SQL, R, NoSQL, Shell, HTML, CSS, Java, Shell Scripting, VBA
- **Frameworks & Tools:** NumPy, Pandas, Matplotlib, Git, GitHub, Docker, Statistics, PySpark, Seaborn, KPI, ggplot2
- **Cloud & Data Engineering:** Azure (Data Factory, Synapse, Data Lake, Blob Storage, Databricks, Azure SQL, Cosmos DB, Functions), AWS, Docker, Apache Spark, Microsoft Fabric, Delta file formats
- **Databases:** MySQL, PostgreSQL, MongoDB, Spark SQL, Oracle, ADF, Data Warehouse
- **Data Analysis:** Power BI, Tableau, OpenCV, Orange, Data Bricks, MS Excel, Google Analytics, Machine Learning, Data Modeling, Data Mapping, Data Extraction, Transformation, Postman, Google Colab, Docker

Professional Experience

Data Analyst Intern, *NSF – Data Science for Energy Transition*

May 2024 – Jun 2024 | USA

- Analyzed large-scale energy transition datasets using Python Pandas, NumPy and SQL, developing predictive models to optimize renewable energy efficiency and improve forecasting accuracy by 20%.
- Built interactive dashboards in Tableau and Power BI, streamlining energy usage monitoring and reducing reporting time by 30% through automated data workflows.
- Automated data preprocessing workflows using Python and SQL, reducing manual data cleaning time by 40% and ensuring high data accuracy

Research Assistant, *University of Houston*

Jan 2024 – May 2024 | USA

- Conducted in-depth data analysis using Python and R, resulting improvement in the accuracy of predictive models for ongoing research projects in data analytics.
- Developed automated data cleaning processes leveraging Pandas and NumPy, reducing manual processing time by 40%, which significantly boosted research productivity.
- Partnered with research teams to analyze findings and contribute to peer-reviewed publications, strengthening the project's impact and academic recognition

Data Analyst, *Infosys Pvt. Ltd*

Jul 2021 – Dec 2022 | India

- Developed an InfoArchive-based application using OpenText products, Core Java, and Python, ensuring compliant archiving and enhancing accessibility of structured and unstructured data.
- Automated project prioritization workflows by creating a data-driven tool in Microsoft Excel and SQL, reducing manual effort by 40% and improving operational efficiency.
- Designed and optimized SQL-based data pipelines and interactive Power BI dashboards, delivering real-time business insights and enabling a 20% reduction in decision-making time for stakeholders.
- Integrated REST APIs to enhance backend data accessibility for web applications, improving response time and optimizing data retrieval processes.
- Conducted data quality analysis and implemented data validation techniques, improving accuracy and consistency across reporting and analytics systems.

Projects

IMDb Top 1000 Movies Dataset

Aug 2023 – Dec 2023

- Analyzed IMDb Top 1000 Movies dataset using Tableau to identify key trends and patterns, resulting in improved insights on box office performance across themes.
- Established interactive dashboards that visualized movie ratings, release years, and themes, increasing user engagement by 35%.

Seoul Bike Sharing Prediction

Jan 2023 – May 2023

- Designed and implemented machine learning models, achieving improvement in bike count predictions by employing advanced algorithms and optimizing model parameters.
- Enhanced model performance by conducting hyperparameter optimization through grid search and random search, resulting in a 20% reduction in prediction errors.

Scalable Azure Data Pipeline for Real-Time E-Commerce Analytics

- Designed and implemented a scalable data transformation pipeline, processing over 1 TB of e-commerce data using Azure Data Factory, Azure Databricks, and Azure Synapse Analytics and conducted real-time data ingestion, reducing data latency by 80%.
- Integrated MongoDB for data enrichment, processing, and cleaning records to improve data quality and usability.
- Configured Azure Synapse to create dedicated and serverless SQL pools, improving query performance by 40%

Certifications & Publications

- Tableau Desktop Specialist Certification – Tableau, 2025
- Microsoft Certified : Azure Fundamentals
- Distributed and Parallelized Image Encryption Using DES Techniques, IRJMETs, June 2020 ([Link](#))